

Modeling the associations between L2 grit, foreign language enjoyment, and student engagement among Chinese EFL English-major learners: A control-value theory perspective

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ABSTRACT

Learning a new language is an intricate and long-term endeavor, fraught with numerous possible obstacles, which needs learners' engagement. Concurrent with the positive psychology of language learners, it is crucial to delve into the role of learners' positive personality features and emotions (e.g., grit, positive emotions) in their engagement. However, there is a scarcity of research on the mediating role of foreign language enjoyment (FLE) in the relationship between English-major students' L2 grit and their engagement in second language (L2) learning processes. By utilizing the control-value theory and structural equation modeling approach, this study seeks to explore the impact of L2 grit components (i.e., the consistency of interest and perseverance of effort) and FLE on the engagement of Chinese undergraduates ($N = 217$) majoring in English in an EFL context. The results indicated that 1) perseverance of effort and the FLE had a significantly positive and direct influence on student engagement; 2) perseverance of effort had a notably positive and direct effect on FLE, while consistency of interest did not affect learners' FLE; 3) consistency of interest had a slightly positive and direct effect on student engagement; and 4) perseverance of effort had a positive and indirect influence on student engagement, partially mediated by FLE, whereas FLE did not mediate the link between consistency of interest and student engagement. These results offer educational insights for improving students' engagement in the language learning process.

1. Introduction

It is widely acknowledged that students' academic performance is closely associated with their engagement in learning-related activities (Derakhshan & Yin, 2024; Fredricks, 2011; Hodge et al., 2018; Li et al., 2023b). Student engagement refers to learners' investment of effort and time in participating in various tasks, interests, and cognitive processes related to learning activities, as well as their commitment to the attainment of learning objectives (Fredricks et al., 2004; Reeve, 2012). Engagement has been identified as a pivotal factor influencing students' academic achievement (Hiver et al., 2024). A plethora of factors have been found to have potential predictive effects on student engagement, including contextual elements (e.g., teacher enthusiasm, teacher support, community environment) (Dewaele & Li, 2021; Khajavy et al., 2018; Sadoughi & Hejazi, 2023), psychological aspects (e.g., enjoyment, anxiety,

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self-efficacy, self-regulation, beliefs) (Derakhshan & Azari Noughabi, 2024; Dewaele et al., 2023; Dewaele & MacIntyre, 2014, 2016; Liu, Darvin, & Ma, 2024), and personality trait-like characteristics and motivational factors (e.g., grit, mindsets, motivation) (Derakhshan et al., 2022; Feng & Papi, 2020; Khajavy et al., 2021). English language learning is a sustained and prolonged learning endeavour. The attainment of significant academic success in English language learning necessitates not only momentary enjoyment but also unwavering perseverance to sustain this linguistic learning journey (Li & Dewaele, 2021; Liu, Zou, Soyoo, & Chiu, 2025; MacIntyre & Khajavy, 2021).

As positive psychology has gained prominence recently, a growing number of researchers and scholars have turned their attention to investigating personality-trait factors (e.g., grit) and positive emotions (e.g., enjoyment) due to their roles in forecasting academic engagement and success (Dewaele & MacIntyre, 2014; Li, 2023; Piniel & Albert, 2018; Tsang & Dewaele, 2024). There was a body of prior studies that have examined the role of grit in language learning (Lee, 2022; Wei et al., 2019). It is worth noting that the domain-general grit contains two sub-construct domain-specific grits (i.e., perseverance of effort (POE) and consistency of interest (COI) (Duckworth & Quinn, 2009). POE is characterized by sustained perseverance and enthusiasm for achieving long-term objectives (Lee & Lee, 2021; Muenks et al., 2017; Zhao & Wang, 2023). COI refers to the ability to maintain consistent interest in the face of failure or adversity (Azari Noughabi & Ghasemi, 2024; MacIntyre & Khajavy, 2021; Zhao & Wang, 2023).

While existing research has extensively documented how personality-trait factors, especially L2 grit, motivate dedicated English learning engagement (Derakhshan & Fathi, 2023; Sadoughi & Hejazi, 2023; Sun et al., 2024) and how positive emotional factors, particularly FLE, foster student engagement (Tsang & Dewaele, 2024; Wang & Hui, 2024), there is a relative dearth of studies that simultaneously consider the predictive roles of L2 grit served as a powerful motive and FLE functioned as a salient and complex affective anchor in academic engagement in English learning activities. Furthermore, of particular interest is L2 domain-specific grit (i.e., POE and COI), a personality-trait variable that has demonstrated promise in empowering academic engagement by fostering learners' persistent efforts and sustaining consistent interests (Li & Yuan, 2024; Sun et al., 2024). This research gap in understanding how two dimensions of L2 grit (i.e., POE and COI) negotiate FLE to engage in learning activities are particularly significant in L2 English learning context, where successful English learners must participate in their perseverance and efforts in sustaining English learning activities in a proactive way (Derakhshan & Fathi, 2023; Usher et al., 2018). Finally, there is a need to conduct further studies on emotions of learners from countries that do not fully align with the WEIRD (Western, Educated, Industrialized, Rich, and Democratic) criteria, as suggested by Dewaele et al. (2023). While China has made significant progress in industrialization and education, its political system and cultural values differ from those of Western democratic societies (Liu & Xie, 2023). As such, the present study aims to contribute to a more comprehensive understanding of how L2 grit and FLE vary in response to factors like diverse learning settings, educational experiences, and career aspirations (Botes et al., 2024) by examining how L2 English students in China negotiate their personality-trait factors (i.e., L2 grit) and modulate their positive emotions (i.e., enjoyment) to engage with English learning. Understanding how L2 grit contributes to this emotional foundation could provide crucial insights for developing more effective pedagogical interventions that facilitate student engagement in English learning environments.

2. Literature review

2.1. Control-value theory

The Control-Value Theory (CVT) offers an extensive framework for examining the precursors and outcomes of emotions encountered in academic and performance-related contexts (Li et al., 2023a, 2023b; Pekrun, 2006). According to CVT, achievement-related emotions influence learning and academic performance through various cognitive processes, including working memory, self-regulation, and metacognition (Shao et al., 2019; Tsang & Dewaele, 2024). The theory categorizes emotions using a three-dimensional model: valence (positive/negative), psychological arousal (activation/deactivation), and object focus (activity-related, outcome-prospective, or outcome-retrospective focus) (Pekrun, 2006). Within this framework, FLE is identified as a positive, activating, and activity-related emotion in the context of L2 learning. Recently, the application of CVT has expanded from general educational psychology to specific L2 educational psychology (Derakhshan & Yin, 2024; Li et al., 2023b; Wang & Kruk, 2024). CVT conceptualizes emotions as interconnected psychological processes, where affective, cognitive, and motivational components are integral (Shao et al., 2019). Regarding the CVT, achievement emotions (e.g., enjoyment, boredom, anxiety) are linked to control-value appraisals, such as control, intrinsic, and attainment value assessments (Li et al., 2023a).

2.2. Grit and foreign language enjoyment

General grit was first presented as a complex construct within the field of positive psychology (Duckworth & Quinn, 2009). L2 grit is recognized as a trait-like personality characteristic that corresponds to intrinsic motivation, facilitating superior academic performance and fostering a steadfast commitment to overcoming obstacles (Azari Noughabi & Ghasemi, 2024; Botes et al., 2024; Derakhshan et al., 2022; Yang et al., 2024). L2 grit consists of two sub-constructs, namely COI and POE (e.g., Lee & Lee, 2021; Li & Dewaele, 2021). However, a large amount of prior research has overlooked an in-depth exploration of the L2 grit structure, often treating it as a unidimensional construct within the proposed models (Botes et al., 2024; Derakhshan & Fathi, 2023; Ebn-Abbasi et al., 2024). This has resulted in an incomplete understanding of the models of domain-specific grit (MacIntyre & Khajavy, 2021). The study of grit in the context of language learning remains relatively underexplored (Feng & Papi, 2020), highlighting the potential for valuable insights in the current research.

A series of empirical studies have already investigated the dual-factor structure of grit. In a study involving ethnic minority L2

Chinese language learners ($N = 504$), Zhao et al. (2023) found that a first-order, two-factor model of grit yielded superior goodness-of-fit indices compared to a one-factor model. This suggested that COI and POE should be examined separately in relation to other affective variables (e.g., FLE). Numerous studies have reported that POE measures outperformed those of COI. For instance, Lee (2022) investigated the relationships among L2 grit, classroom enjoyment, and willingness to communicate (WTC) across three groups—junior high school, senior high school, and tertiary students ($N = 647$). The findings indicated that POE significantly predicted L2 WTC, while COI did not (Lee, 2022). Similarly, Usher et al. (2018) found that POE was a stronger predictor of academic performance, compared to COI, suggesting that COI and POE should be treated as distinct constructs in further investigations. It is noteworthy that these findings align with Zhao et al. (2023), who observed that both COI and POE had significant and positive impacts on FLE and English learning achievement. However, POE showed a stronger effect power, compared with COI, in predicting emotions (i.e., FLE and boredom) and English language achievement (Zhao & Wang, 2023). These differences might ascribe variations to language learners and learning contexts, warranting further exploration into the structure of grit.

FLE describes the pleasure and fulfillment that learners feel while participating in foreign language learning activities (Dewaele et al., 2023; Dewaele & MacIntyre, 2014). Enjoyment is the most frequently encountered form of pleasure among learners (Piniel & Albert, 2018) and has numerous beneficial effects on university students (Dewaele et al., 2016; Dewaele & Li, 2021). In the context of Chinese EFL learning, L2 grit has been found to positively influence English language performance, with FLE mediating the relationship between L2 grit and foreign language performance (FLP) (Wei et al., 2019). Within the context of Iranian EFL learners, Derakhshan and Azari Noughabi (2024) investigated the impact of EFL learners' FLE, foreign language peace of mind, and psychological capital on academic engagement through a self-determination perspective among a sample of 200 participants. Their findings indicated that FLE exerted the most significant positive influence on academic engagement.

2.3. Student engagement and foreign language enjoyment

Student engagement, a multifaceted construct that includes behavioral, cognitive, affective, and social components, is vital in shaping academic success or failure (Hiver et al., 2024; Reeve, 2012). Behavioral engagement specifically relates to learners' active involvement in educational tasks and is regarded as critical for obtaining favorable academic results (Fredricks, 2011). Affective engagement involves learners' emotional responses, both positive and negative, towards learning activities or tasks (Fredricks et al., 2004; Li, 2023). Cognitive engagement pertains to the degree to which learners invest thoughtful consideration and are willing to devote time and effort to complete a given task (Li et al., 2023b; Mystkowska-Wiertelak, 2022). Social engagement refers to the degree to which learners actively seek external assistance when engaging in learning activities, such as reaching out to teachers, peers, friends, or family members (Svalberg, 2009).

A wealth of research has highlighted the impact of FLE on the academic learning of students (Dewaele & MacIntyre, 2014; Lee & Lee, 2021; Liu, Zou, Soyoof, & Chiu, 2025). For example, Dewaele and Li (2021) discovered that the combination of enjoyment and boredom among 2002 Chinese EFL university students influenced the connection between teacher enthusiasm and social-behavioral learning engagement. In another study exploring the links between L2 grit, FLE, self-efficacy, and learners' online engagement, FLE was found to have a positive effect on the online engagement of Iranian EFL learners (Derakhshan & Fathi, 2023). Wu and Yu (2023) investigated the relationship between students' personality traits and e-learning among a sample of 1004 university students during the COVID-19 pandemic. Their research indicated that enjoyment could serve as a mediator between personality traits (such as extroversion, openness to new experiences, agreeableness, and conscientiousness) and e-learning engagement. In a recent cross-cultural study by Derakhshan and Yin (2024), the relationship between positive activating emotions and academic engagement was examined among Chinese and Iranian EFL learners, revealing that FLE could positively influence academic engagement in both groups. Likewise, Wang and Hui (2024) found that enjoyment had a significant positive effect on online English learning engagement.

2.4. Grit, foreign language enjoyment, and student engagement

An increasing number of studies have offered empirical evidence on the connection between L2 grit and student engagement. For instance, Sadoughi and Hejazi (2023) found that both external influences (e.g., teacher support) and internal factors (e.g., growth language mindset) positively predicted Iranian EFL learners' academic engagement. Similarly, Derakhshan and Azari Noughabi's (2024) study uncovered that FLE can positively influence academic engagement among Iranian EFL learners. Additionally, it is reported that the students who demonstrate greater perseverance in overcoming difficulties and maintaining sustained interest can significantly enhance their academic engagement (Sadoughi & Hejazi, 2023). Moreover, Derakhshan and Fathi (2023) identified that FLE positively influenced online learning engagement, and online learning self-efficacy mediated the interaction between L2 grit and online engagement among 578 Iranian EFL learners. In the context of L2 writing, the study of Li and Yuan (2024) indicated that COI significantly impacted cognitive engagement, while POE influenced affective, cognitive, and behavioural engagement in various ways. Significantly, emotional and behavioral engagement acted as intermediaries in the link between L2 perseverance and English writing outcomes. Hodge et al. (2018) found that engagement could function as a beneficial mediator in the relationship between determination and academic success, with COI and POE playing a crucial role in shaping engagement at Australian universities.

However, to the best of our knowledge, existing research on the predictive role of EFL learners' trait-level personality features in their engagement is quite limited, with relatively little attention given to L2 grit (Azari Noughabi & Ghasemi, 2024; Ebn-Abbasi et al., 2024). Due to the significance of comprehending learner engagement language learning, it is evident that more studies are needed to explore the relationships between L2 grit components, FLE, and student engagement. Additionally, earlier studies have highlighted the mediating function of FLE in leading teacher enthusiasm to increased levels of learner engagement in language classes (Dewaele & Li,

2021). Furthermore, the mediating influence of FLE has been confirmed among Chinese middle and high school students studying English (Wei et al., 2019). Nevertheless, whether FLE can act as a mediator in the relationship between L2 grit and learner engagement, especially when L2 grit is viewed as a dual-component model, remains an open question that requires further investigation.

2.5. The present study

In answer to the particular appeal to combine L2 grit and positive emotions (i.e., FLE) to further progress the field, this study aimed to investigate how domain-specific L2 grit and FLE can predict student engagement in English language learning activities (Dewaele & Li, 2021; Sadoughi & Hejazi, 2023; Wang & Hui, 2024). In particular, the current study examined the mediating role of FLE.

Reviewing the previous research suggests that the majority of studies have predominantly focused on non-English major EFL learners, while studies targeting English major EFL learners remain relatively scarce (Wang & Kruk, 2024). Given that English majors incorporate English learning as an integral part of their academic studies, it is closely tied to their future careers in English-related fields (e.g., English teachers, translators, interpreters, or business professionals) to some extent (Ngo et al., 2017; Zou & Xu, 2017). The motivation for learning English differs between English majors and non-English majors, with English majors exhibiting stronger intrinsic motivation and feeling less obligated to learn English compared to their non-English major counterparts (Liu, Darvin, & Ma, 2024). This implies that students with intrinsic motivation are more inclined to dedicate time and energy to learning English (Liu, Zhang, & Zhang, 2024; Sun et al., 2024). Consequently, it is valuable to investigate the connections between L2 grit, FLE, and student engagement, particularly among English major EFL learners. The research questions are as follows:

RQ1: What is the underlying structure of L2 grit in the context of Chinese EFL English-major university students?

RQ2: To what extent do L2 grit and FLE play predictive roles in Chinese EFL English-major students' engagement?

RQ3: To what extent does FLE mediate the relationship between L2 grit and student engagement among Chinese EFL English-major students?

Additionally, COI and POE were included separately in the model to further examine which aspect of L2 grit serves as the most effective predictor in L2 English learning among Chinese English majors. As independent factors, POE, COI, and FLE were hypothesized to predict student engagement (STE). The seven hypotheses, rooted in the related literature, outlined below, have been put forward.

- H1. COI makes a positive impact on FLE;
- H2. POE makes a positive impact on FLE;
- H3. FLE makes a positive impact on STE;
- H4. COI makes a positive impact on STE;
- H5. POE makes a positive impact on STE;
- H6. FLE mediates the relationship between COI and STE;
- H7. FLE mediates the relationship between POE and STE.

3. Methodology

3.1. Participants

Situated within the Chinese university context, this research recruited participants majoring in English from two universities located in Central China. All participants were recruited through online channels. Adopting the convenience sampling technique for the sample selection (Creswell & Creswell, 2017), 429 students majored in English at the two universities. The male-to-female gender ratio was approximately 1:10. The research team first circulated a research e-poster that contained an online questionnaire QR code via WeChat, a prominent Chinese social media platform, to 14 English teachers who lecture English-major students' curriculum and introduced our research aims and the gist of the online questionnaire to these English instructors who assisted in disseminating the questionnaire to their students between August and October 2023. Subsequently, the instructors communicated with the students to ensure they understood the purpose of the study when participating in this study. Participants were required to meet one essential criterion: current enrolment as an English major student at a Chinese university.

A total of 222 undergraduate English majors from China voluntarily participated in the study, with 217 valid responses retained after data cleaning. Before the study began, participants were asked to complete an online informed consent form. After conducting data screening and cleaning, a total of 217 valid questionnaire responses were retained. Out of the 217 participants, 22 were male (10.1 %) and 195 were female (89.9 %). The ages of the participants ranged from 18 to 22 years. The sample consisted of first-year students (N = 80, 36.8 %), second-year students (N = 88, 40.5 %), third-year students (N = 21, 9.6 %), and fourth-year students (N = 28, 12.9 %).

3.2. Instruments

The data collection instrument was a composited questionnaire, including demographical information items, an adapted *L2 grit scale*, an original *Foreign language enjoyment scale*, and an original *Student engagement scale* (see more details in [Appendix A](#)). There were two parts structured in the survey. Demographic details in the initial part were collected from respondents, such as gender, age, and academic year. Thirty-four statements in the subsequent part, derived from three established scales, aimed at measuring respondents' L2 grit, FLE, and student engagement. Participants were asked to rate these statements on a 5-point Likert scale, with options ranging from 1 (strongly disagree) to 5 (strongly agree).

3.2.1. L2 grit scale

To measure the participants' L2 grit, we modified and employed the general Grit scale originally created and validated by [Duckworth and Quinn \(2009\)](#), which was empirically validated by prior research in the Chinese EFL context (L2 domain-specific grit scale) ([Liu & Wang, 2021](#); [Zhao & Wang, 2023](#)). The 8-item L2 grit scale encompasses two dimensions: Consistency of Interest (COI) and Perseverance of Effort (POE) (with four items in each dimension). To better suit the English learning context, we adapted the statements of items to specify the English learning context. Specifically, we added "in English learning" to nine items of the Grit scale respectively. For example, one statement for the COI dimension was modified into "I often set a goal but later choose to pursue a different one in English learning." One item for the POE dimension was revised into "I am a hard worker in learning English." Overall, we made minor adjustments to the expression of each item, retaining the original meaning of the items. Additionally, the interest dimension includes four reverse-scored items (e.g., "New ideas and projects sometimes distract me from previous ones in learning English"), while the perseverance dimension consists of four items, such as "Setbacks don't discourage me in learning English." The items are rated on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree). In this study, the reliability coefficients for L2 grit and its two sub-dimensions (COI and POE) were .77 (total scale), .71 (interest), and .78 (perseverance), indicating acceptable internal consistency ([Field, 2013](#)).

3.2.2. Foreign language enjoyment scale

This research employed the 11-item FLE scale created by [Li et al. \(2018\)](#) to examine the predictive and mediating functions of FLE in the language acquisition process. The scale is divided into three dimensions: FLE-private (e.g., "I've learnt interesting things."), FLE-teacher (e.g., "The teacher is encouraging."), and FLE-atmosphere (e.g., "There is a good atmosphere."). The responses were captured using a 5-point Likert scale, with options ranging from 1 (strongly disagree) to 5 (strongly agree). An example item begins with "I don't get bored in learning English." In this study, the overall reliability coefficients for FLE and its three sub-dimensions (FLE-student, FLE-teacher, FLE-atmosphere) were .86 (total scale), .85 (FLE-student), .86 (FLE-teacher), and .86 (FLE-atmosphere), indicating that the scale demonstrates excellent reliability.

3.2.3. Student engagement scale

To measure the participants' engagement, the *Student Engagement Scale*, developed and validated by [Li et al. \(2023a\)](#), was utilized. This scale was designed to examine students' levels of L2 engagement (15 items) and its role within the underlying mechanisms. The L2 student engagement scale was selected due to its successful application in the Chinese EFL learning context, making it well-suited to the Chinese language learning system ([Li et al., 2023b](#)). The original scale comprised four subscales measuring different constructs of engagement: behavioural (five items), affective (three items), cognitive (four items), and social engagement (three items). Behavioural engagement was assessed through five statements, such as "If I do not understand what is being taught in English class, I will try my best until I figure it out" ([Li et al., 2023b](#)). Affective engagement was evaluated using three statements, including "I always wish the English classes could be longer when they are finished" ([Li et al., 2023b](#)). Cognitive engagement was measured via four statements, such as "I listen carefully in English class in order not to miss anything important" ([Li et al., 2023b](#)). Social engagement included three reverse-scored statements, such as "I hope not to be called on for any questions in English classes" ([Li et al., 2023b](#)). This instrument demonstrated strong internal consistency (Cronbach's $\alpha = .90$) and validity in the Chinese context ([Li et al., 2023b](#)). The items of this scale were rated on a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). In the present study, the reliability coefficients for the total scale and the individual constructs were .92 (total scale), .83 (behavioural engagement), .82 (cognitive engagement), .80 (affective engagement), and .81 (social engagement), indicating excellent internal consistency.

3.3. Data collection

The survey items were presented utilizing bilingual versions in both English and Chinese to enhance participants' comprehension during the questionnaire completion process. As the *FLE scale* and the *Student engagement scale* were pre-validated in both languages, their bilingual content was directly incorporated into this study. Conversely, the original *L2 grit Scale* was formulated in English, prompting the involvement of a doctoral candidate specializing in translation studies from the United Kingdom. This doctoral candidate, whose native language is Chinese, has learned English as his foreign language and achieved high levels of proficiency in both languages. He also holds a master's degree in translation studies." He was tasked with translating the scale (from English to Chinese) and also conducted a back-translation (from Chinese to English), achieving a remarkable back-translation accuracy rate of 98 %. The survey, employed as a tool for quantitative data gathering, was conducted via an online survey platform Wenjuanxing (<https://www.wjx.cn/>) ([Dörnyei & Taguchi, 2009](#)). It commenced with a required question inquiring whether participants agreed to take part in the study. Only individuals who responded with "Yes" were allowed to proceed to the main questionnaire. The gathered data were

examined using Structural Equation Modeling (SEM) through AMOS 26.0, adhering to a five-step procedure (Collier, 2020; Kline, 2023). The data collection process involved the following steps: first, participants completed the web-based consent forms before data collection began. Second, they were asked to complete online questionnaires consisting of two sections (Section 1: demographic information; Section 2: surveys evaluating L2 grit, FLE, and student engagement).

3.4. Data analysis

Initially, a systematic data screening process was carried out to detect outliers, missing data, and invalid entries (e.g., surveys completed in under 2 min). This phase also involved assessing the normality of the dataset by examining the skewness and kurtosis of each item. Next, descriptive statistics were generated to calculate the means and standard deviations for the survey items. In the third step, Cronbach's alpha was computed for each factor to determine reliability. The fourth step involved validating the instruments through Confirmatory Factor Analysis (CFA). Finally, based on the reliability and validity outcomes, SEM and mediation analysis were performed to test the proposed model, investigating the relationships among L2 grit, FLE, and student engagement.

4. Results

4.1. Construct of L2 grit

In this research, CFA was employed to verify the factor structure of L2 grit by testing three different models: 1) a first-order single-factor model, 2) a first-order two-factor model, and 3) a second-order two-factor model (refer to Table 1). The first-order two-factor model of L2 grit showed a better fit in comparison to the other two models, as indicated by the *Goodness-of-fit* indices. Factor loadings ranged between .53 and .93. As a result, this study separately analyzed the predictive roles of POE and COI (refer to Fig. 1).

4.2. Descriptive statistics and preliminary analyses

Table 2 displays the means (M), standard deviations (SD), skewness, and kurtosis for L2 grit (encompassing both COI and POE), FLE, and four dimensions of student engagement (i.e., behavioral, affective, cognitive, and social engagement). The normality of the data distribution was assessed according to the recommended criteria for skewness (within |3|) and kurtosis (within |8|) as outlined by Kline (2005), indicating that the data met the assumptions of normality. Descriptive statistics further revealed that the means and deviations for each construct fell within the range of the 5-point scale, with means ranging from 2.71 to 3.57 and standard deviations ranging from .89 to 1.07. Among the constructs, COI had the lowest mean, while FLE had the highest mean score. In Table 2, all Cronbach's α coefficients were above .7.

4.3. CFA of variables and the measurement model

We examined the CFA of each variable before testing the measurement model. The CFA of L2 grit (i.e., COI and POE) have been reported in the 4.1 section. As the initial indices of the CFA of FLE did not fully meet the recommended values by Collier (2020), the CFA of FLE was recalculated after removing observed variables (i.e., F1, F5, F8-F11) based on the modification indices (refer to Table 3). In Table 3, all five indices of FLE fell within the recommended benchmark range, indicating that the measurement model and the structural model exhibits a good fit with the dataset. Additionally, regarding CFA of STE, after deleting the observed variable (i.e., BE5), the CFA of STE can fully achieve the benchmark value suggested by Collier (2020) (see Table 3). Tables 4 and 5 summarize the findings of the measurement model with respect to item reliability, convergent validity, and discriminant validity. As recommended by Hair et al. (2006), two key reliability indicators were reported: Cronbach's alpha ($\alpha > .7$) and composite reliability ($CR > .6$). Similarly, validity analysis includes two validity metrics: convergent validity ($AVE > .5$) and discriminant validity ($HTMT < .9$), which was assessed using the Heterotrait-Monotrait Ratio of Correlations (HTMT) criterion as proposed by Henseler et al. (2015).

Furthermore, CR for each construct exceeded the recommended minimum of .6 (Hair et al., 2006). Every factor loading surpassed the .5 threshold, and the average variance extracted (AVE) values were all greater than .5 (Fornell & Larcker, 1981), suggesting robust internal consistency within the constructs.

As illustrated in Table 5, the HTMT values for the seven factors generally fell below the acceptable threshold of .9, which remains within acceptable boundaries. This suggests that the seven factors are statistically distinguishable from one another (Henseler et al., 2015).

Table 1
Goodness-of-fit indices for three models of L2 grit (N = 217).

Model	χ^2	df	χ^2/df	CFI	GFI	TLI	NFI	RMSEA	SRMR
First-order one-factor	139.27	9	15.47	.71	.82	.51	.70	.26	.13
First-order two-factor	13.99	8	1.75	.99	.98	.98	.97	.06	.05
Second-order two-factor	no value								
Cutoff values (Kline, 2023)	The smaller, the better	/	<3	>.90	>.90	>.90	>.90	<.10	<.08

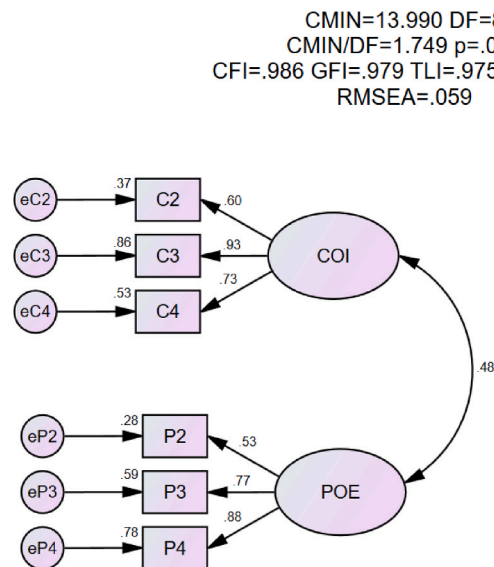


Fig. 1. The first-order two-factors model of L2 grit.

Table 2
Descriptive statistics.

Constructs	Mean	SD	Skewness (< 3)	Kurtosis (< 8)	Cronbach's a (>.7)
COI	2.71	1.01	.49	-.11	.71
POE	3.26	.89	-.33	-.19	.78
FLE	3.57	.90	-.63	.69	.86
BE	3.34	.93	-.48	-.03	.83
AE	3.31	.95	-.32	-.12	.80
CE	3.48	.92	-.63	.27	.82
SE	3.17	1.07	-.25	-.65	.81

Note: COI = Consistency of Interest; POE = Perseverance of Effort; STE = Student Engagement; BE = Behavioural Engagement; Affective Engagement; CE = Cognitive Engagement; SE = Social Engagement.

Table 3
The CFA of FLE and student engagement.

CFA	χ^2/df	CFI	TLI	NFI	RMSEA
FLE	2.03	.99	.976	.98	.07
STE	2.48	.94	.918	.90	.08

4.4. Structural model

The structural model was assessed using a variety of fit indices, such as the chi-square statistic relative to degrees of freedom (χ^2/df), Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), Incremental Fit Index (IFI), Parsimony Normed Fit Index (PNFI), Parsimony Goodness-of-Fit Index (PGFI), Root Mean Square Error of Approximation (RMSEA), and Standardized Root Mean Square Residual (SRMR). As shown in Table 6, all eight indices met the recommended thresholds (Hu & Bentler, 1999), indicating that the proposed structural model is acceptable. It is noteworthy that this study incorporated a second-order variable, student engagement, necessitating an analysis of its component weights (see Table 7).

The path coefficients for the structural model were detailed in Table 8 and outlined in Fig. 2. The results indicated that POE (H2: POE $\rightarrow$ FLE, $\beta = .60$, $p < .001$, $t\text{-value} = 6.41$) positively affected FLE, supporting H2. Additionally, FLE significantly and positively impacted Student Engagement (H3: FLE $\rightarrow$ STE, $\beta = .40$, $p < .001$, $t\text{-value} = 5.25$), supporting H3. COI positively influenced Student Engagement (H4: COI $\rightarrow$ STE, $\beta = .13$, $p = .005$, $t\text{-value} = 2.81$), supporting H4. POE had a positive effect on Student Engagement (H5: POE $\rightarrow$ STE, $\beta = .26$, $p < .001$, $t\text{-value} = 3.42$), supporting H5. However, COI did not directly influence FLE (H1: COI $\rightarrow$ FLE, $\beta = .04$, $p = .602$, $t\text{-value} = .52$), leading to the rejection of H1. Concerning the R^2 values, COI and POE collectively explained 43 % of the variance in FLE, whereas the model accounted for 70 % of the total variance in Student Engagement (refer to Fig. 2). These R^2 values displayed the model's notable explanatory power and the strong effect sizes of the statistically significant associations, as proposed by Plonsky and Oswald (2014).

Table 4
Reliability and convergent validity of the constructs in the measurement model.

Constructs	Items	Factor loadings (>.5)	CR (>.6)	AVE (>.5)
Consistency of interest	C2	.60	.80	.58
	C3	.93		
	C4	.73		
Perseverance of effort	P2	.53	.78	.55
	P3	.77		
	P4	.88		
Foreign language enjoyment	F2	.80	.84	.52
	F3	.79		
	F4	.58		
	F6	.86		
Behavioural engagement	F7	.50	.84	.56
	BE1	.71		
	BE2	.77		
	BE3	.70		
Affective engagement	BE4	.81	.81	.59
	AE1	.75		
	AE2	.74		
Cognitive engagement	AE3	.81	.83	.55
	CE1	.63		
	CE2	.71		
	CE3	.81		
Social engagement	CE4	.80	.83	.62
	SE1	.85		
	SE2	.60		
	SE3	.89		

Table 5
The discriminant validity of the measurement model.

	HTMT (<.9)						
	COI	POE	FLE	BE	AE	CE	SE
COI	–						
POE	.54	–					
FLE	.33	.67	–				
BE	.45	.69	.70	–			
AE	.49	.80	.77	.88	–		
CE	.54	.79	.66	.85	.86	–	
SE	.47	.48	.48	.60	.65	.66	–

Note: COI = Consistency of Interest; POE = Perseverance of Effort; STE = Student Engagement; BE = Behavioural Engagement; Affective Engagement; CE = Cognitive Engagement; SE = Social Engagement.

Table 6
The proposed measurement model and SEM model fit indices.

	χ^2/df	CFI	TLI	IFI	PNFI	PGFI	RMSEA	SRMR
The measurement model	1.93	.92	.90	.92	.72	/	.07	.06
The structural model	1.92	.92	.91	.92	.74	.69	.07	.06
Cutoff values (Kline, 2023)	<3	>.90	>.90	>.90	>.50	>.50	<.08	<.10
Evaluated	Good	Good	Good	Good	Good	Good	Good	Good

Table 7
Weight distribution of student engagement in second-order models.

Second-order variable	First-order variables	<i>p</i>	Impact weight
Student engagement	STE → BE	***	.88
	STE → AE	***	.98
	STE → CE	***	.94
	STE → SE	***	.64

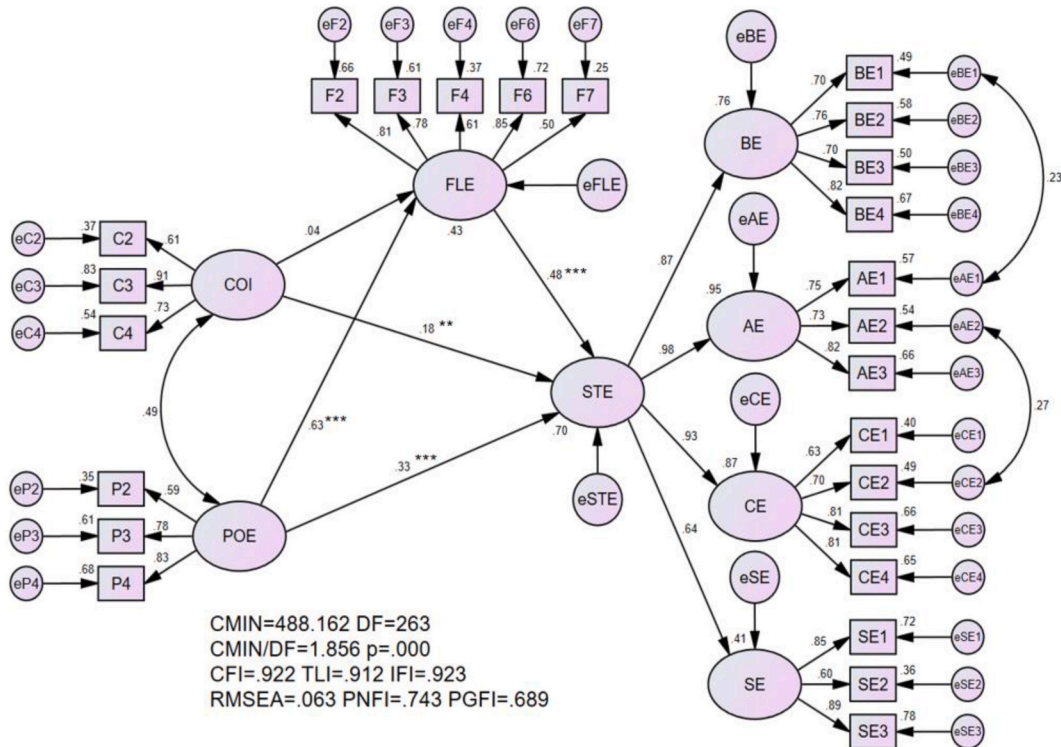
Note: ****p* < .001. STE = Student engagement; BE = Behavioural engagement; AE = Affective engagement; CE = Cognitive engagement; SE = Social engagement.

Table 8

Path coefficients of the structural model.

Hypothesis	Path	β	S.E.	C.R. (t-value)	p	Results
H1	COI $\rightarrow$ FLE	.04	.07	.52	.602	Unsupported
H2	POE $\rightarrow$ FLE	.60	.09	6.41	$p < .001^{***}$	Supported
H3	FLE $\rightarrow$ STE	.40	.08	5.25	$p < .001^{***}$	Supported
H4	COI $\rightarrow$ STE	.13	.05	2.81	.005**	Supported
H5	POE $\rightarrow$ STE	.26	.08	3.42	$p < .001^{***}$	Supported

Note: COI = Consistency of Interest; POE = Perseverance of Effort; STE = Student Engagement. *** $p < .001$; ** $p < .01$; * $p < .05$.

**Fig. 2.** The finalized structural model.

Note: R^2 : FLE = 43 %, Student Engagement = 70 %; COI = Consistency of Interest; POE = Perseverance of Effort; STE = Student Engagement; BE = Behavioural Engagement; AE = Affective Engagement; CE = Cognitive Engagement; SE = Social Engagement.

4.5. The mediating effects

In order to assess the mediating effect of FLE on the relationship between COI, POE, and student engagement, a bootstrap analysis with a sample size of two thousand and a 95 percent confidence interval was conducted to estimate the mediating effect of foreign language enjoyment. Results were reported in Table 9. Table 9 presented the indirect effects, 95 percent confidence intervals, and p -values for the structural model, indicating that POE had a positive and indirect influence on student engagement through FLE (H7: POE $\rightarrow$ FLE $\rightarrow$ STE, $\beta = .24$, $p = .006$), demonstrating a partial mediating effect. However, FLE did not mediate the relationship between COI and student engagement (H6: COI $\rightarrow$ FLE $\rightarrow$ STE, $\beta = .01$, $p = .683$), which did not support H6.

Table 9

The mediating effect results.

Path	β	p (Two-tailed Significance)	95 % Confidence Interval		Results
			Lower bound	Upper bound	
H6: COI $\rightarrow$ FLE $\rightarrow$ STE	.015	.683	-.052	.081	Unsupported
H7: POE $\rightarrow$ FLE $\rightarrow$ STE	.238	.006**	.094	.389	Partial mediation

Note: $p < .01^{**}$. COI = Consistency of Interest; POE = Perseverance of Effort; STE = Student Engagement; BE = Behavioural Engagement; AE = Affective Engagement; CE = Cognitive Engagement; SE = Social Engagement.

5. Discussion

This research marks a novel attempt to explore the predictive influence of L2 grit's two-factor components and FLE on student engagement in the context of Chinese EFL. It is among the first attempts to validate the factor structure of L2 grit among English majors, exploring the mediating role of FLE in the relationships between L2 grit components and student engagement from a holistic perspective. This research responds to previous calls for the combined assessment of grit and emotions (Derakhshan et al., 2022; Zhao & Wang, 2023). From the perspective of positive psychology, this study represents a pioneering effort to explore how L2 grit and FLE shape student engagement among Chinese EFL English major students. From a theoretical standpoint, the findings align with the principles of positive psychology, which propose that positive trait-like personality features (i.e., L2 grit) are recognized as pivotal cornerstones in facilitating language learners' progress (Azari Noughabi & Ghasemi, 2024; Derakhshan et al., 2022; Derakhshan & Fathi, 2023; Feng & Papi, 2020; MacIntyre & Khajavy, 2021). Furthermore, these findings lend additional credibility to the CVT (Pekrun, 2006), suggesting that a heightened awareness and utilization of positive emotions and personality features can enhance learners' engagement in language learning, thereby improving language proficiency (Muenks et al., 2017; Sadoughi & Hejazi, 2023; Wang & Hui, 2024).

Regarding RQ1, the findings revealed that L2 grit was best represented as a first-order, two-factor construct, highlighting the significance of its two interconnected components: COI and POE. Therefore, in order to understand the relationship between L2 grit and other factors in more detail, it is suggested that COI and POE be used as separate dimensions when assessing their importance in language learning. This approach helps prevent the dilution of their distinct influences (Hodge et al., 2018; Muenks et al., 2017; Sadoughi & Hejazi, 2023). The results of the present study, derived from participants who are Chinese English majors within an EFL context, are consistent with Zhao and Wang (2023), whose study focused on ethnic minority students.

Concerning RQ2, the three major findings demonstrated the direct and predictive roles of L2 grit and FLE in student engagement. First, this study has found that two major components of L2 grit (i.e., COI and POE) positively and significantly impacted student engagement in English learning environments by means of direct ways, which aligned with prior studies on how learners' persistent efforts and consistent interests foster their academic engagement (Hodge et al., 2018; Muenks et al., 2017; Sadoughi & Hejazi, 2023). These insights were reflected in the strong predictive effects of COI and POE on student engagement respectively (H4: COI $\rightarrow$ STE, $\beta = .13$, $p = .005$, $t\text{-value} = 2.81$; H5: POE $\rightarrow$ STE, $\beta = .26$, $p < .001$, $t\text{-value} = 3.42$). On this basis, it was indicated that the COI and POE served as stable personality trait forces for L2 English learners' proactive and prolonged engagement in English learning. This finding is not only consistent with the study of Sadoughi and Hejazi (2023) in terms of L2 grit's capability in predicting academic engagement among Iranian EFL learners but also provides evidence for the pivotal role of personality-trait factors in conditioning online language learning behaviors and engagement (Derakhshan & Fathi, 2023; Sun et al., 2024).

In addition, the present study unveiled that POE positively, significantly, and directly predicted FLE (H2: POE $\rightarrow$ FLE, $\beta = .60$, $p < .001$, $t\text{-value} = 6.41$), while COI did not impact FLE (H1: COI $\rightarrow$ FLE, $\beta = .04$, $p = .602$, $t\text{-value} = .52$). This insight is reminiscent of the case developed by Lee (2022) that 647 Korean EFL students from junior high schools, senior high schools, and tertiaries with a high degree of L2 grit that focused perseverant efforts may be more likely to engage with communication practices in English because the intrinsic and continuous personality-trait motives often successfully provide sustainable drivers and support for their autonomous learning behaviours (Muenks et al., 2017), whereas COI did not exert a significant influence on learners' engagement and willingness to interact in English in the study of Lee (2022). In contrast, our present finding diverges from the research made by Zhao and Wang (2023), who found that both COI and POE significantly and positively predicted FLE. This discrepancy may be attributed to distinct EFL learner populations, among which substantial individual differences may exist, particularly in terms of educational resources and curriculum requirements of English learning. Chinese university English-major students may face demanding and heavy academic pressure stemming from curriculum or coursework that made them realize that effortful persistence, more than interest consistency, can enable them to overcome academic difficulties, make progress, and experience more enjoyment in English learning (Zou & Xu, 2017). It should also be noted that COI and POE contributed 43 % variance in explaining FLE in the SEM model, which revealed that L2 grit, particularly POE, played a central and crucial role in inspiring enjoyment in the English as a lingua franca world.

Finally, the current study revealed that a sense of FLE enabled greater engagement in English learning, an observation that was reflected in the significant and direct predictive impact of FLE on student engagement in a manner of a positive way (H3: FLE $\rightarrow$ STE, $\beta = .40$, $p < .001$, $t\text{-value} = 5.25$) in the SEM model. It echoed the prior studies that EFL students with higher levels of language learning enjoyment displayed a greater willingness to engage in English learning, as internal positive emotions prompted autonomous learning behaviours (Derakhshan & Azari Noughabi, 2024; Derakhshan & Yin, 2024; Dewaele & Li, 2021; Li, 2023; Liu, Darvin, & Ma, 2024b; Wang & Hui, 2024; Wu & Yu, 2023). Notably, in the present research, we also found that the effects of L2 grit and FLE on student engagement are neither linear nor discrete. Not only did FLE exert a direct and significant role in student engagement, but also it played a mediating role between L2 grit and student engagement, a complicated association that was elaborated below.

Addressing RQ3, the key findings disclosed that FLE mediated the influence that a personality-trait force (i.e., POE) had on student engagement in English learning activities, which was corroborated by the quantitative result (H7: POE $\rightarrow$ FLE $\rightarrow$ STE, $\beta = .24$, $p = .006$), while FLE did not exert a mediation effect on the relationship between COI and student engagement (H6: COI $\rightarrow$ FLE $\rightarrow$ STE, $\beta = .01$, $p = .683$). Consistent with the findings of prior studies (Derakhshan & Fathi, 2023; Li, 2023; Wu & Yu, 2023) with regard to the complexity and malleability of language enjoyment in relation to L2 learners' grit, this study revealed that persistent efforts of Chinese EFL university students empowered the sense of enjoyment that bolstered their engagement with English learning activities. Notably, the findings related to FLE and student engagement further support the application of CVT in the SLA (Pekrun, 2006), showing the sustainable positive affections facilitated learning engagement and enhanced attainment.

Furthermore, our findings revealed that substantial differential effects of personality-trait forces and a positive affective variable on

student engagement, collectively accounting for 70 % of its variance, which indicated that L2 grit and FLE were recognized as significant components and driving motives in the language learning process, ultimately promoting learners' engagement and academic achievement due to the strong explanatory effect powers of personality-trait factors and positive emotions. It was noteworthy that although COI positively influenced student engagement, interest alone was less effective in enhancing student engagement through FLE, implying COI had a weaker impact on student engagement than POE. There were some potential reasons illustrated below. On the one side, in the Chinese cultural values and educational norms, Chinese EFL learners may be more predisposed to derive academic engagement from steadfast and effortful persistence in the face of adversity rather than interest-driven enjoyment. This inclination can be ascribed to the cultural emphasis placed on diligence, hard work, and perseverance (Li & Yuan, 2024; Liu & Wang, 2021). On the other side, the underlying instability and superficiality of interest in English learning could restrict the extent to which interest alone can sustain academic engagement, thereby rendering persistent efforts more crucial (Feng & Papi, 2020). Thus, the finding suggested that perseverant effort maximized its impact on student engagement through the mediation of positive emotions. It was posited that heightened levels of enjoyment can maintain or even amplify perseverance, thereby nurturing greater student engagement (Derakhshan & Fathi, 2023).

6. Conclusions, implications, and limitations

This study examined the impact of L2 grit and FLE on student engagement with L2 learning among Chinese English major students. In particular, the present study explored the mediating role of FLE in the relationships between EFL learners' COI and POE and their engagement. The findings suggest that L2 grit, encompassing both COI and POE, is a crucial element of one's academic achievement or success that significantly contributes to increasing academic engagement. Notably, POE is essential in fostering enjoyment in language learning, while COI contributes minimally to this aspect. Furthermore, the connection between POE and student engagement emphasizes the mediating role of FLE, shedding light on the interplay between perseverance and enjoyment in the language learning process. These results suggest that future research should focus on integrating personality traits with various emotions to further understand their combined impact on student engagement.

Building upon the present research insights, several implications are proposed for future pedagogical practices. **First**, educators and language teachers should recognize the power of L2 grit as a source of academic tenacity and motivation for sustaining language learning engagement. To cultivate the development of L2 grit, they should create a classroom culture that values perseverance and praises students' efforts. Specifically, when a student successfully completes a challenging task, the teacher could say something like, "I noticed how much effort you put into preparing for this task. You did it very well, and you have improved a lot!". In addition, language teachers could organize goal-setting activities that promote students' L2 grit and maintain long-term language learning goals, thereby reinforcing academic engagement. For example, teachers organize students to create a language-learning vision board where they set specific, measurable, and time-bound goals for the semester or year and encourage students to regularly review and reflect on their phased progress towards these goals, and celebrate their achievements. **Second**, given the positive and direct effect of L2 grit on language enjoyment, language teachers could invite successful language learners to share their stories. These learners could highlight how they found joy in the language learning process despite the challenges they faced, emphasizing the role of perseverance and effort in their success. This encouragement can motivate students to sustain their learning tenacity, thereby evoking learning enjoyment. **Third**, given the positive and direct role of FLE on student engagement, language teachers may incorporate and introduce game-based language learning games and activities that are both educational and enjoyable. For instance, teachers organize students to play word-guessing games like "Pictionary" to improve vocabulary and communication skills. Employing gamification teaching activities enables to creation of a relaxed, enjoyable, engaging language learning environment, which increases enjoyment and strengthens proactive participation in language learning. **Finally**, concerning the indirect and positive impact of L2 grit on academic engagement through the catalyst FLE, practitioners and teachers can create enjoyable language learning practices that cater to students' learning interests, needs, and styles, which could help students view persistent effort as a rewarding and enjoyable endeavour, thus sustaining their positive emotion and reinforcing learning behaviours. For instance, teachers implement a reader's theatre activity where students select a scene from a play or a book, assign roles for students, and encourage students to perform the scene in English. This approach combines language learning with elements of self-accomplishment and self-fulfilment, teacher support and encouragement, welcoming and inclusive learning milieu, which helps students expand language learning enjoyment, thereby promoting active engagement.

While this study offers valuable insights, it contains several limitations. Firstly, this research employed a cross-sectional design to explore the predictive roles of L2 grit components (COI and POE) and FLE on student engagement, which overlooks the influence of students' transient emotions on task engagement during different language learning tasks. Future research could adopt a longitudinal approach to investigate the relationships among L2 grit, FLE, and task engagement across various task contexts. Using a growth curve modeling approach can provide promising results regarding the change in the relationships between the main variables in the long run. Such a longitudinal study would allow for a comparative analysis of changes over time, capturing the dynamic interactions between personality traits and emotions among students. Second, the sample in this study was limited to a small group of Chinese university students majoring in English from two institutions located in central China, which may constrain the generalizability of the findings to other EFL learner populations in China, such as non-English majors, learners in different educational settings, or those from other geographical regions within the country. Consequently, future studies should involve participants from a wider range of backgrounds to investigate the underlying learning mechanisms, as the predictive influence of constructs such as grit and FLE might differ across various populations, educational contexts, language learning objectives, and learning environments and consider large-scale sample size to enhance the generalizability and robustness of the findings. Third, demographics have not been studied in this research. The

gender of the participants may play a moderating role in the associations among L2 grit, FLE, and student engagement. Concerning the majority of participants who were female, further studies with more balanced samples are needed. Furthermore, it is recommended that future studies combine L2 grit with a range of emotions (e.g., hope, pride, boredom, shyness) to explore their various effects on student engagement. This approach would provide a more thorough understanding of the connection between L2 grit and emotions, illuminating how these elements jointly impact student engagement. Finally, future studies may consider employing mixed-methods approaches to better understand the nuanced and complicated interplay mechanisms through which persistent efforts and consistent interests impact diverse facets of language affective factors and engagement.

CRedit authorship contribution statement

Minlin Zou: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mostafa Azari Noughabi:** Writing – review & editing, Validation, Supervision. **Caixia Peng:** Writing – review & editing, Resources, Investigation, Data curation.

Data availability statement

The raw data can be made available upon request.

Consent for publication

Informed consent for publication was obtained from all individual participants included in the study.

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No potential conflict of interest was reported by the authors. There is no potential conflict of interest associated with this research study.

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Appendix A. Supplementary data

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